HARDWARE BRING-UP REPORT

Rotocast Controller V4

First power-up of the new controller board. Two design faults made the machine unusable as built; both are now understood, reworked and verified. A third issue limits how hard the motors can be driven and needs a decision before the next batch.

BOARD	FAB ORDER	UNITS TESTED	FIRMWARE
`002_Rotocast_ControllerV2`	`2963065A_Y819`	2 — both identical	MicroPython 1.29.0

STATUS AT A GLANCE

SUBSYSTEM	RESULT	NOTE
RP2040, flash, USB, 3V3 rail	Good	No issues at any point
Display panel & 13 flex joints	Good	Panel answers over SPI; all joints sound
Display backlight	Fault 2	Q3 pinout — reworked
Main button SW3	Fault 1	Footprint — reworked
Encoder and push switch	Good	Firmware pin map was wrong, corrected
Addressable LED	Good	
TMC2209 drivers & UART bus	Good	Answer at addresses 0 and 1
Driver thermals	Limiting	See section 3 — caps useful torque
Both motors	Good	1–25 cage rpm, ratio verified
Fan output	Good	Fan needs ≥55% duty to start
12 V input path	Good	Use a regulated supply — see section 5

1 — MAIN BUTTON IS PERMANENTLY GROUNDED

SW3 BLOCKING REWORKED

The footprint uses two legs that are common inside the switch

SW3 is an **HX B3F-4055-Y** (LCSC C36498965), a 12 mm four-leg tactile switch. Its four legs form two rails: the two legs on one row are permanently joined inside the body, as are the two on the other row, and pressing the button bridges the rails.

The board wires the signal to pad 2 and ground to pad 1 — **both on the same row, so both on the same rail.** The button line is therefore tied to ground whether or not anyone presses it. Pads 3 and 4, the other rail, are left unconnected.

FROM THE PRODUCTION GERBER NET ATTRIBUTES

```
SW3 pad 1  (175.77, -106.71)  → GND
SW3 pad 2  (160.77, -106.71)  → signal (R9 / R10 node)
SW3 pad 3  (175.77, -101.71)  → N/C
SW3 pad 4  (160.77, -101.71)  → N/C
```

ON THE BENCH

```
Signal node to ground  0.2 Ω, released and pressed
GPIO25 held low against its ~50 kΩ internal pull-up, indefinitely
GPIO25 can still drive the line high → a resistive tie, consistent with R10's
100 Ω
```

The consequence in firmware is that the input reads "pressed" forever, so the button can never be seen to change state. Both boards behaved identically.

REWORK APPLIED

One switch leg lifted clear of its pad and a short jumper run to the unused pad 5 mm away, putting signal and ground on different rails. Verified: open when released, ~0 Ω when pressed, and the firmware now reads the button correctly.

NEXT REVISION

Map the symbol's two pins to legs from *different* rails — pads 2 and 4, or 1 and 3. Worth checking the same footprint isn't reused elsewhere.

Q3 BLOCKING REWORKED

The schematic symbol's pin order doesn't match a SOT-23 transistor

Q3's symbol numbers the device **collector-base-emitter** on pins 1-2-3. A real SOT-23 NPN — S8050, and the fitted part — is **base-emitter-collector**. Fitted to this footprint, the drive signal lands on the emitter, the LED path on the base, and ground on the collector. It cannot switch the backlight in any state, so the screen was dark on both boards.

DIODE TEST ACROSS Q3, IN CIRCUIT

pad 1 → pad 2	**2.8 V**	pad 1 → pad 3	**1.4 V**
pad 2 → pad 1	OL	pad 2 → pad 3	OL

Conduction flows **out of pad 1** to both other pads — the signature of a base. Pad 1 is the pad the board uses for the collector.

SUPPORTING MEASUREMENTS

- Drive node (pad 2) **0 V** with the backlight commanded both on and off
- GPIO13 verified driving its output high — the MCU side is fine
- Backlight LED itself healthy: 2.5 V forward, one direction only
- Top marking on the fitted part read **X0PL**, not the J3Y of a standard S8050

REWORK APPLIED

Q3 removed and its collector pad bridged to the ground pad. The backlight now runs permanently, with current set by R31's 22 Ω at roughly 16 mA. Software dimming is lost, which this machine does not need.

NEXT REVISION

Correct the symbol to base-emitter-collector. Separately, **R21** (10 kΩ from base to +3V3) should pull *down* to ground — as drawn, the backlight defaults on at power-up before firmware runs.

J2 / J3 NEEDS A DECISION

No cooling path for two drivers running near 1 A

The TMC2209 modules overheat well below the current the machine wants. Measured with both motors running at 10 cage rpm, heatsinks fitted, no forced airflow, each run started below 120 °C:

Current sweep, 45 s per setting

SETTING	COIL CURRENT	RESULT
8/31	≈ 0.50 A rms	Stayed under 120 °C throughout
12/31	≈ 0.72 A rms	One driver over 120 °C at 35 s
16/31	≈ 0.94 A rms	Over 120 °C in 5 s, thermal shutdown by 20 s

Chopper mode matters as much as current. At 24 rpm, StealthChop crossed 120 °C after 28 seconds and sat between 120 and 143 °C indefinitely; SpreadCycle at the same current *shed* heat from a hot start and dropped back below 120 °C. Firmware now uses SpreadCycle above 6 cage rpm and runs at 0.50 A.

Why this needs your input

The machine currently runs on bare motors with no cage fitted. The previous machine reached 25 rpm at the cage under real load, so once the cage and a mould are on, 0.50 A may not be enough — and raising it without cooling is exactly what put both drivers into shutdown. **What current and what drivers did the old machine use?** If it managed 25 rpm loaded without thermal trouble, its drivers were either better cooled or lower-impedance than these modules, and that should shape the next board.

Options, cheapest first: airflow over J2/J3 — the board already drives a fan, it just isn't aimed at the electronics; a better thermal path under the modules; or driver modules with more copper. Worth designing a fan mount or a copper pour into the next revision.

4 — TRACE CLEARANCES ON THE BUTTON NET

/BTN ADVISORY

Two spots sit below normal fab minimums

Measured from the production gerbers by rasterising the copper layers at 5 µm and growing the button nets until they touch ground copper:

MINIMUM CLEARANCE TO GROUND POUR

```
/BTN net   →  0.065 mm  near the via beside C19
switch-node net  →  0.100 mm  beside R10
```

Neither caused a fault on these two boards, and the via's own annular clearance is fine. But 0.065 mm is under the 0.0889 mm most fabs hold as standard, so it's a yield risk rather than a design error. Worth opening both to 0.2 mm.

5 — SUPPLY QUALITY RESETS THE DRIVERS

12 V DOCUMENT IT

A sagging supply silently reverts the drivers to their pin defaults

Running from a 3S LiPo, both drivers repeatedly lost their UART configuration mid-run. A TMC2209 that browns out comes back taking microstepping from its MS pins and current from its trim pot — and because MS1 also sets the UART address, **the two drivers come back at different microstep resolutions**: 8 and 32 against the 64 the firmware commands.

AFTER A BROWNOUT

```
driver 0   GCONF=0x000101   CHOPCONF=0x15010053  → 8 microsteps, pot-set current
driver 1   GCONF=0x000101   CHOPCONF=0x13010053  → 32 microsteps, pot-set current
```

CONFIGURED CORRECTLY

```
both   GCONF=0x0001C0   CHOPCONF=0x12000053  → 64 microsteps, UART-set current
```

The visible symptom was one motor trying to run eight times too fast and stalling, which looks exactly like a torque problem and is not one. On a regulated 12 V 5 A supply it stopped completely. Firmware now re-applies the configuration at every start and checks it every two seconds while running, so it recovers on its own — but the machine should be specified with a regulated supply.

6 — SMALLER NOTES

R12 NO ACTION

Marked DNP, and correctly not fitted

The empty pads at the processor's corner pin look like a missed placement on inspection. They are the DNP pull-up on the flash chip-select, exactly as the schematic specifies. Noting it here so nobody else chases it.

Fan WORTH NOTING

The fitted fan will not start below about 55% duty

Firmware maps its range onto 55–100% and gives it a short full-power kick to break it loose. Worth either specifying a fan with a lower start threshold or recording this in the BOM notes.

Encoder CORRECTED

Pin map in the original test firmware was wrong

The board wires the encoder A to GP6, B to GP5 and the push switch to GP4; the supplied `test.py` used GP5, GP4 and GP3, so the knob press did nothing and rotation was read from the wrong pins. The board is correct — the code was fixed.

J2 / J3 CONFIRMED GOOD

Driver addressing works as designed

M1 answers at UART address 0 and M2 at address 1 via the MS1 straps, and the single-wire bus through R6 is sound — verified by loopback and by both drivers replying with version 0x21.

7 — WHERE THE MACHINE STANDS

With both reworks in place the controller has run continuously for six minutes at up to 25 cage rpm with no thermal warnings, no stalling and no lost driver settings. A full functional pass — controls, ramping, ratio, timer, stop behaviour, supply-cycle recovery and settings persistence — passed end to end.

New controller firmware replaces the two-year-old Arduino code, keeping its controls (cage rpm, ratio for the inside motor, run timer, start/stop button, knob to select and adjust) and adding the fan, the colour display, per-driver temperature readout, an over-temperature stop, and settings that survive a power

cycle. Speeds on screen are cage rpm through the 20T:80T reduction, matching the old machine's scale exactly.

Outstanding decisions for the next board

ITEM	RECOMMENDATION
SW3 footprint	Signal and ground on legs from different internal rails
Q3 symbol	Base-emitter-collector pin numbering for SOT-23
R21	Pull the transistor base down to ground, not up to 3V3
Driver cooling	Fan mount or improved thermal path at J2/J3 — decides the torque ceiling
Button net clearances	Open 0.065 mm and 0.100 mm spots to 0.2 mm
Supply	Specify regulated 12 V; battery sag resets the drivers

Prepared from bench measurements on two boards, the production gerbers for 2963065A_Y819, and the Rev 2 schematic. Register values, clearance figures and thermal timings are quoted as measured.